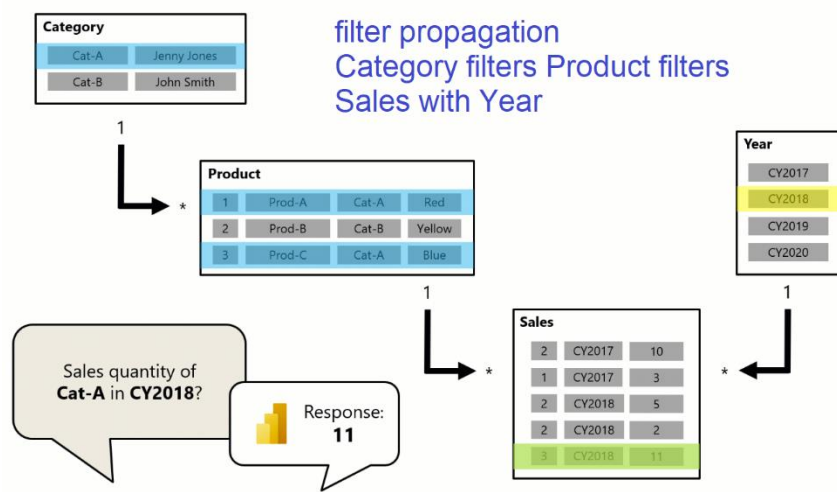

PBI Relations & Filters

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Discussion

1. [DAX sample model - DAX | Microsoft Learn](#)
2. *Examples in this article can be used with the sample Adventure Works DW 2020 Power BI Desktop model. To get the model, see [DAX sample model](#)*
3. [Model relationships in Power BI Desktop - Power BI | Microsoft Learn](#)
4. [CALCULATE function \(DAX\) - DAX | Microsoft Learn](#)
5. [CALCULATETABLE function \(DAX\) - DAX | Microsoft Learn](#)
6. [USERELATIONSHIP function \(DAX\) - DAX | Microsoft Learn](#)
 - a. Cannot be used when row level security is defined for the table in which the measure is included.
 - b. Example: Override the default, active, relationship between InternetSales and DateTime tables.
 - c. To calculate the sum of internet sales and allow slicing by ShippingDate instead of the traditional OrderDate,
 - d. Create measure, [InternetSales by ShippingDate]
 - e. = CALCULATE(SUM(InternetSales[SalesAmount]),
USERELATIONSHIP(InternetSales[ShippingDate], DateTime[Date]))
7. [Tutorial: From Dimensional Model to Stunning Report in Power BI Desktop - Power BI | Microsoft Learn](#)



DAX

Copy

```
CALCULATE(<expression>[, <filter1> [, <filter2> [, ...]])
```

Term	Definition
expression	The expression to be evaluated.
filter1, filter2,...	(Optional) Boolean expressions or table expressions that defines filters, or filter modifier functions.

The expression used as the first parameter is essentially the same as a measure.

Filters can be:

- Boolean filter expressions
- Table filter expressions
- Filter modification functions

Filter modifier functions

Filter modifier functions allow you to do more than simply add filters.

They provide you with additional control when modifying filter context.

Function	Purpose
REMOVEFILTERS	Remove all filters, or filters from one or more columns of a table, or from all columns of a single table.
ALL ¹ , ALLEXCEPT, ALLNOBLANKROW	Remove filters from one or more columns, or from all columns of a single table.
KEEPFILTERS	Add filter without removing existing filters on the same columns.
USERELATIONSHIP 	Engage an inactive relationship between related columns, in which case the active relationship will automatically become inactive.
CROSSFILTER	Modify filter direction (from both to single, or from single to both) or disable a relationship.

```

DAX
Blue Revenue =
CALCULATE(
SUM(Sales[Sales Amount]),
'Product'[Color] = "Blue"
)

```

Category	Sales Amount	Blue Revenue
Accessories	\$1,272,057.89	\$165,406.62
Bikes	\$94,620,526.21	\$8,374,313.88
Clothing	\$2,117,613.45	\$259,488.37
Components	\$11,799,076.66	\$803,642.10
Total	\$109,809,274.20	\$9,602,850.97

The CALCULATE function evaluates the sum of the **Sales** table **Sales Amount** column in a modified filter context.

A new filter is added to the **Product** table **Color** column—or, the filter overwrites any filter that's already applied to the column.

Adventureworks Example

In Power BI with AdventureWorks data, USERELATIONSHIP activates an inactive relationship (e.g., Shipping Date) to calculate metrics (e.g., Shipped Amount) alongside an active one (Order Date). Use a DAX measure with CALCULATE to temporarily override the model's active relationship, allowing sales reporting based on alternate dates.

Scenario: Sales by Order Date vs. Ship Date

In the AdventureWorks model, the FactInternetSales table often links to the DimDate table via OrderDateKey (Active) and ShipDateKey (Inactive).

1. **Active Relationship:** DimDate[DateKey]

FactInternetSales[OrderDateKey]

2. **Inactive Relationship:** DimDate[DateKey]

FactInternetSales[ShipDateKey]

DAX Measure Example

To calculate the total sales amount based on the **Shipping Date** rather than the order date, create this measure:

dax

Sales Amount by Ship Date =

```
CALCULATE(  
    SUM(FactInternetSales[SalesAmount]),  
    USERELATIONSHIP(FactInternetSales[ShipDateKey], 'DimDate'[DateKey])  
)
```

Key Details

- **CALCULATE:** The USERELATIONSHIP function must be used within CALCULATE or CALCULATETABLE.
- **Override:** The measure forces the relationship, regardless of whether it is active or inactive in the model view.
- **Result:** This measure can be used in visuals to show sales by Shipping Date, while a standard SUM measure will still show Sales by Order Date.
- **Blank Values:** Note that if an order has not shipped, USERELATIONSHIP will return a BLANK value for the ship date.

Reference

[Exam PL-300: Microsoft Power BI Data Analyst - Certifications | Microsoft Learn](#)

1. [Course PL-300T00--A: Microsoft Power BI Data Analyst - Training | Microsoft Learn](#)
2. <https://github.com/MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst>

Analyzing & Large Report Example

1. [**Data, Probability and Statistics \(mathsisfun.com\)](#)
2. [Ship INSIGHT \(Container Ship\) Registered in Marshall Is - Vessel details, Current position and Voyage information - IMO 9790907, MMSI 538007806, Call Sign V7ZZ9 | AIS Marine Traffic](#)
3. [Filter a report using query string parameters in the URL - Power BI | Microsoft Learn](#)
4. [**Data, Probability and Statistics \(mathsisfun.com\)](#)
5. endlist